IW Project Documentation

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1 What was done

Project 2: Weather app

User can search for locations: → Search Field

User can use his/her location GPS-coordinates (Geolocation API): → `currentLocation()` using: Google Geolocation API

At least two data/forecast providers are used (completely different data sources): Yes

At least three data/forecast providers are used: OpenWeatherMap, Open-Meteo, MeteoBlue

User sees the current weather at a specific location: → `searchBtn` and funtion `fetchWeatherData("city", linkCurentw, city, null, unitDefault);` shown with `current(data)`

User sees the forecast for the next 24 hour, hourly based: → 8x3 hour format `fetchWeatherData("city", linkForw, city, null, unitDefault);` shown with `hourlyfor(data)`

User sees the forecast for the next 7 days: → in chart `fetchWeatherData("sevencity", null, city, null, unitDefault);` shown with `sevendayforcast(data)`

All the weather forecast elements use icons (and numbers) Yes, from: `https://openweathermap.org/img/wn/${icon}.png`

The look and feel of the application reflects the current weather Yes,but only temperature-based: `cold`, `cool`, `warm`, `hot` using: `pagecolor(unit, temp)` to set `body.classList`

User sees simultaneously two forecasts in a graph Yes — OpenWeather vs Open-Meteo Shown in `tempCharts.html` using current location

User has the option to tag some locations as favorites: No database, no login, so saving "favorite" would be little useless in the same session, but I save the previous searches and they are available via `addtopreSerch(city)`

User has an option to switch between Celsius, Fahrenheit, and Kelvin: Yes — top right corner

Application is responsive and can be used on both desktop and mobile environment Yes, it's responsive and also small "animations" on the card's + Different Menu for desktop and mobile: `"Big screen>800px" "800>Medium>600" "600>Mobile"`

2 Tools used - Links: footer of `index.html`

- HTML, CSS, JavaScript
- Bootstrap 5 (I used CSS and set some of my own styles, but some of the styling and example-component's came from there)
- Frappe Charts
- mdboootstrap/weather (ideas for the cards)
- APIs used:
  - OpenWeatherMap
  - Open-Meteo
  - MeteoBlue
- Geolocation API

3 Point's

I implemented all the required features. I would like to recive **40** points

4 AI

AI (GitHub Copilot) was used in the following 2 cases to get some help:

- **Date Time Formatting** I had trouble with getting them into nice format. I started with Googeling it of course, but most of them were for different use cases and i wanted to use different formats, I gave up. what i used originally
  Example Prompt: API returns date as 1760799600, API Doc says its "Time of data forecasted, Unix, UTC" format. Help formatting it?
  Solution **hh:mm** format `scripts/indexscript.js line 240 new Date(item.dt * 1000).toLocaleTimeString([], hour: "2-digit", minute: "2-digit" );`
- **Bootstrap Utilities** I know how to use Bootstrap, but AI helped me choose and combine some of the utility div classes for the card's. To make them look better.
  But all of them can be found here in Docs/utilities
  Example: `class="d-flex justify-content-around mt-3"` or `class="small mb-1 fw-semibold"`